Network Transform: : **CHEAT SHEET**

About

Kaiaulu's graph module unifies a variety of data sources (e.g. git logs, mailing lists) in a consistent object-oriented structure using S3 (R's object-oriented system). This abstraction allows future graph functions to be de-coupled from the data source. Graph functions will also apply the correct transformations depending on the graph object being passed as parameter (similarly to a print() function).

Downloaders

Kaiaulu downloaders allow different raw data to be obtained. For reference, see refreshers cheatsheet: cheatsheets/refresher-cheatsheet.pdf

1. download_jira_issues()
Downloads JIRA data locally as JSON files.
2. ...

Parsers

Parse functions format raw data into structured tables that are reusable across multiple transforms.

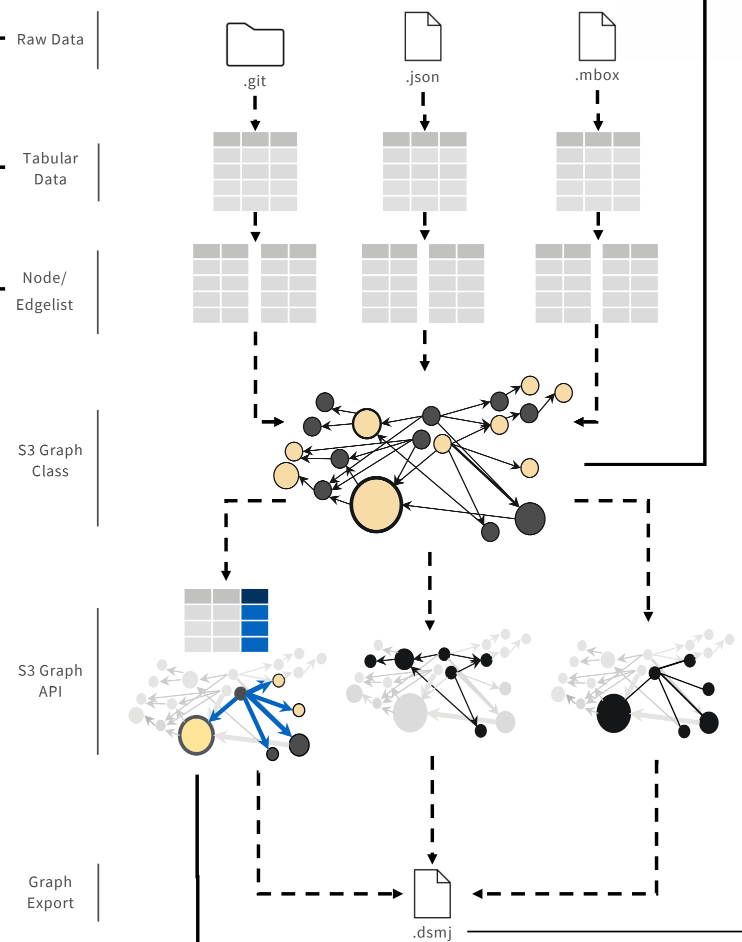
1. parse_gitlog()
Parses a git repository into a commit table.
2. parse_jira_replies()
Parses Jira issue comments into a reply table.
3. ...

Transformers

Transform functions construct S3 graph objects from parsed tables by defining nodes, edges, and relationships.

1. transform_gitlog_to_bipartite_network()
Git Log commit table → bipartite graph.
2. transform_gitlog_to_multimodal_network()
Git Log commit table → multimodal graph.
3. transform_reply_to_bipartite_network()
Reply table → person-issue bipartite graph.
4. ...

Network Transform Pipeline



Graph Constructors

Transformers utilize a graph structure which generalizes different types of graphs, regardless of their data source.

1. model_unimodal_graph()
Represents a unimodal graph.
2. model_multimodal_graph()
Represents a multimodal graph.

S3 Functions

S3 methods execute the correct behavior based on the graph's class, allowing for analyses functions to work regardless of the data's origin.

Examples of S3 Generic Functions:
1. bipartite_graph_projection()
Performs bipartite projection on multimodal graphs.
2. temporal_graph_projection()
Performs temporal projection on multimodal graphs.
3. community_oslom()
Runs OSLOM community detection on a graph.
4. graph_to_dsmj()
Exports graph to a .dsmj file.

Example of S3 Auxiliary Functions:
5. subset_bipartite_from_multimodal()
Slices a bipartite graph from a multimodal graph.
6. get_bidirected_edges()
Returns all mutual edge pairs from a graph.

Custom Graph Weights

Any numeric value — such as sentiment scores or custom weight schemes — can be assigned as edge **weights** without modifying any existing transforms.

1. transform_reply_to_bipartite_network(weight, ...)
2. transform_gitlog_to_bipartite_network(weight, ...)
3. ...

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